



Apocynum venetum leaf aqueous extract inhibits voltage-gated sodium channels of mouse neuroblastoma N2A cells

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ABSTRACT

Ethnopharmacological relevance: *Apocynum venetum* Linn. (Apocynaceae family), also called Luobuma, is a shrub which grows widely in the Xinjiang Autonomous Region of China. Its leaves are used in herbal tea for the treatment of hypertension, anxiety and depression. Animal studies have also shown that *Apocynum venetum* leaf extract (AVLE) also exerts anti-depressant and anti-anxiety activities. The effects of AVLE on neuronal tissues *in vitro* are not fully understood.

Materials and methods: Using the whole-cell voltage-clamp method, we studied the effects of AVLE on ion channels in cultured mouse neuroblastoma N2A cells.

Results: AVLE inhibited voltage-gated inward Na⁺ current in a reversible and concentration-dependent manner (half-inhibitory concentration was 18 µg/ml and maximum inhibition at 100 µg/ml). AVLE specifically promoted steady-state inactivation of Na⁺ channels but did not affect voltage-dependence of activation. The inhibitory effect was not use-dependent and was not affected by 300 µM L-NAME, suggesting that NO was not involved in the action of AVLE in neuronal cells. AVLE also had a mild inhibitory effect on voltage-gated K⁺ channels, but did not affect ATP-sensitive K⁺ channels.

Conclusions: Since voltage-gated Na⁺ and K⁺ channels are associated with neuronal excitability and therefore affect neurotransmission, the modulation of neuronal ion channels by AVLE may exert neuropharmacological effects. In particular, the inhibition of voltage-gated Na⁺ currents by AVLE may in part account for the psychopharmacological effects of this herbal remedy.

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1. Introduction

The leaves of *Apocynum venetum* Linn. (Apocynaceae family) have long been used as traditional herbal tea for the preventive measure against blood stagnation in China and Japan. Recent studies of *Apocynum venetum* leaf extract (AVLE) have demonstrated its general antioxidant effects (Cao et al., 2003; Shirai et al., 2005) and anti-hypertensive effects in spontaneously hypertensive rats possibly through its enhancement of urinary excretion of Na⁺ and K⁺ (Kim et al., 2000). The antihypertensive effects of AVLE have also been attributed to vasorelaxant mechanisms involving endothelium-derived nitric oxide (NO), endothelium-derived hyperpolarizing factor and activation of vascular smooth muscle K⁺ channels (Kwan et al., 2005; Tagawa et al., 2004). A mild

cardiotonic effect in guinea pig heart has been reported for AVLE; such an effect appears to be related to inhibition of phosphodiesterase 3 (Irie et al., 2009).

Leaves of *Apocynum venetum* have also been known to have remarkable psychopharmacological effects (anti-anxiety and anti-depression). AVLE has been shown to exert antidepressant effects in rats (forced swimming test) (Butterweck et al., 2001). A subsequent work by the same group showed that long-term (8-week) AVLE treatment decreased the levels of noradrenaline and dopamine, but not serotonin, in rat hypothalamus, striatum and hippocampus (Butterweck et al., 2003). AVLE has also been reported to have anxiolytic effects in rats (elevated plus maze test); the anxiolytic effect appeared to be mediated by GABA_A receptor and 5-HT_{1A} receptor since it could be blocked by the benzodiazepine antagonist flumazenil or 5-HT_{1A} receptor antagonist WAY-100635 (Grundmann et al., 2007, 2009).

Although there have been several reports about the psychopharmacological effects of AVLE, not much is known about the effects of AVLE on neuronal tissues or neuronal ion channels *in vitro*. A

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recent report demonstrates that AVLE, by inhibiting the activities of pro-apoptotic caspases, could protect rat cortical neurons from injury induced by oxygen and glucose deprivation (Xiang et al., 2010). AVLE could also protect corticosterone-induced apoptotic cell death in PC12 cells (Zheng et al., 2011). Since ionic channels have been implicated in neuroprotection, it would be of interest to examine if AVLE also affected neuronal ionic channels. In this work we investigated whether AVLE affected voltage-gated Na^+ channels and voltage-gated K^+ channels in mouse neuroblastoma N2A cells. We found that AVLE concentration-dependently suppressed voltage-gated Na^+ channels and had a relatively milder effect on voltage-gated K^+ channels.

2. Materials and methods

2.1. Chemicals

AVLE was commercially prepared as a standard water-soluble extract in powdered form under the trade name VENETRON (Tokiwa Phytochemical Company, Tokyo, Japan). Voucher specimens of the leaves of *Apocynum venetum* Linn. were kept in the laboratory of Tokiwa Phytochemical Company. Lidocaine, diazoxide and L-NAME were obtained from Sigma–Aldrich (St. Louis, MO). Glibenclamide was obtained from Calbiochem (San Diego, CA).

2.2. Cell culture

N2A cells were grown at 37 °C in 5% CO_2 in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (Invitrogen, Carlsbad, CA) and penicillin–streptomycin (100 units/ml, 100 $\mu\text{g}/\text{ml}$) (Invitrogen). Rat insulinoma RINm5F cells were cultured at 37 °C in 5% CO_2 in Roswell Park Memorial Institute 1640 medium (Invitrogen) supplemented with 10% fetal bovine serum (Invitrogen) and penicillin–streptomycin (100 units/ml, 100 $\mu\text{g}/\text{ml}$) (Invitrogen).

N2A cells were used because their ion channel properties have been well studied. Depolarization of N2A cells triggered voltage-gated Na^+ currents (Fig. 1A; Leung et al., 2010a). We did not detect voltage-gated Ca^{2+} currents in N2A cells using the voltage-clamp method (Leung et al., 2010a), despite that a very weak Ca^{2+} signal stimulated by 70 mM KCl could be observed using microfluorimetric imaging and fura-2 as dye (Leung et al., 2011). This suggests a very low expression of voltage-gated Ca^{2+} channels which run down rapidly once the whole-cell configuration is formed. The Na^+ currents could be completely inhibited by 1 μM tetrodotoxin and 100 μM lidocaine (tonic block) (Leung et al., 2010a,b).

2.3. Electrophysiology

Electrophysiological experiments were performed as described in previous reports (Chou et al., 2009; Leung et al., 2010a,b). N2A cells were voltage-clamped in the whole-cell mode. Thin borosilicate glass capillary tubes (OD 1.5 mm, ID 1.10 mm, Sutter Instrument, Novato, CA) were prepared with a micropipette puller (P-87, Sutter Instrument), and then fire-polished by a microforge (Narishige Instruments, Inc., Sarasota, FL). The extracellular solution was composed of (mM): 140 NaCl, 4 KCl, 1 MgCl_2 , 2 CaCl_2 , 10 HEPES (pH 7.4 adjusted with NaOH). For the measurement of Kv currents, the intracellular solution was composed of (mM): 140 KCl, 1 MgCl_2 , 1 EGTA, 10 HEPES, and 5 MgATP (pH 7.25 adjusted with KOH). For the measurement of K_{ATP} currents, the concentration of MgATP in the above intracellular solution was reduced to 2 mM. For the measurements of Na^+ currents, the intracellular solution had (mM): 120 CsCl; 20 TEA-Cl; 8 NaCl; 1 MgCl_2 ; 1 EGTA; 10 HEPES and 5 MgATP (pH 7.25 adjusted with CsOH). Pipette resistance ranged

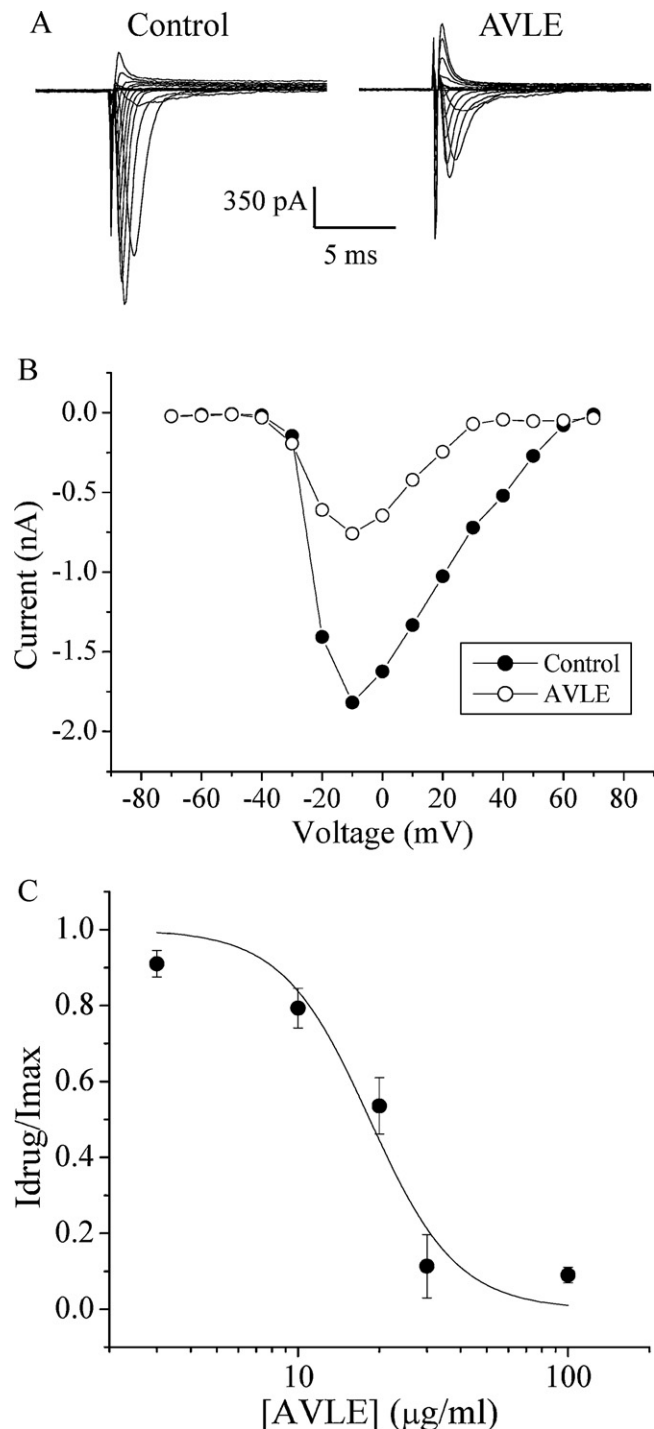


Fig. 1. Voltage-gated Na^+ currents in N2A cells were inhibited concentration-dependently by AVLE. (A) Representative traces of sodium currents triggered by depolarization (-70 to $+70$ mV in 10 mV increments at a holding potential of -70 mV) in the absence and presence of 20 $\mu\text{g}/\text{ml}$ AVLE. (B) I–V relationship of results obtained from (A). Similar results were obtained in at least 5 more experiments. (C) Concentration-response curve of Na^+ current inhibition by AVLE. Results represent mean $\pm$ SEM from 3–7 cells of each group.

from 3 to 7 $\text{M}\Omega$. The currents were recorded with an EPC-10 amplifier with a Pulse 8.60 acquisition software and analyzed by a Pulsefit 8.60 software (HEKA Elektronik, Lambrecht, Germany). Data were filtered and sampled at 2 kHz and 10 kHz, respectively. As soon as a whole-cell configuration was established, the membrane potential was held at -70 mV and subject to increasing depolarization pulses (-70 to $+70$ mV, 10-mV increments) to obtain current–voltage rela-

tion curves, or to repetitive depolarization pulses (–10 mV, 10-s intervals) to examine the inhibitory effects by AVLE. All experiments were conducted at room temperature (~24 °C). To construct the activation curves, Na⁺ currents were stimulated with increasing depolarization, and conductance (*G*) was calculated as:

$$G = \frac{I}{V - V_r}$$

where $V_r = (RT/zF) \ln(Na_o/Na_i)$. *V* is the applied voltage, *V_r* is the reversal potential of Na⁺, *I* is the current, *R* is the universal gas constant, *T* is the temperature, *z* is the ion valency (+1 in this case) and *F* is the Faraday constant. *Na_o* and *Na_i* are, respectively, extracellular and intracellular Na⁺ concentrations. The Boltzmann equation was used to fit the data for voltage-dependence of activation:

$$\frac{G}{G_{\max}} = \frac{1}{1 + \exp[(V_{1/2} - V)/k]}$$

where *G_{max}* is the maximum conductance, *V_{1/2}* is the half-maximal activation potential, and *k* is the slope factor.

To construct the steady-state inactivation curve, a dual-pulse protocol was used in which a test pulse step of –10 mV was preceded by a pre-pulse of 2 s of different potentials. The test pulse currents (*I*) are normalized to the largest test pulse current (*I_{max}*) and plotted against the pre-pulse voltages. The Boltzmann equation was used to fit the data for steady-state inactivation:

$$\frac{I}{I_{\max}} = \frac{1}{1 + \exp[(V - V_{1/2})/k]}$$

where *V_{1/2}* is the half-maximal inactivation potential, and *k* is the slope factor.

2.4. Statistical analysis

Data are represented as means ± SEM. Unpaired or paired student's *t* test was used where appropriate to compare two groups, and *p* < 0.05 was considered to be of statistical significance.

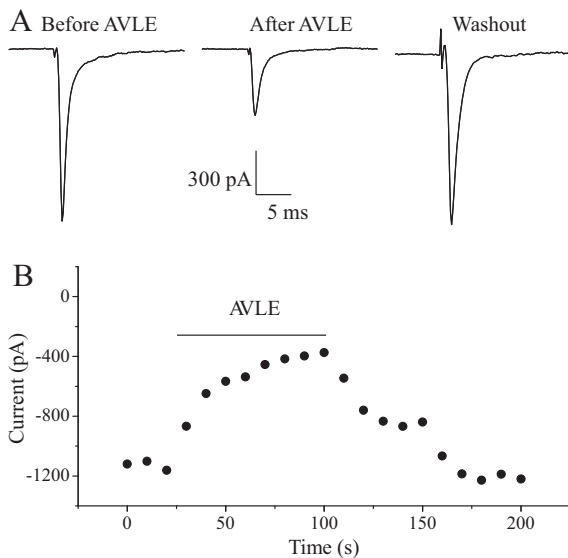


Fig. 2. Block of voltage-gated Na⁺ currents in N2A cells by AVLE was rapid and readily reversed upon washout. (A) Currents were triggered by –10 mV stimulations every 10 s during which 20 μg/ml AVLE was added and then washed out. The typical traces shown are currents before and after AVLE addition, and after washout of AVLE. (B) The peak currents are plotted against time. Similar results were obtained in 2 more experiments.

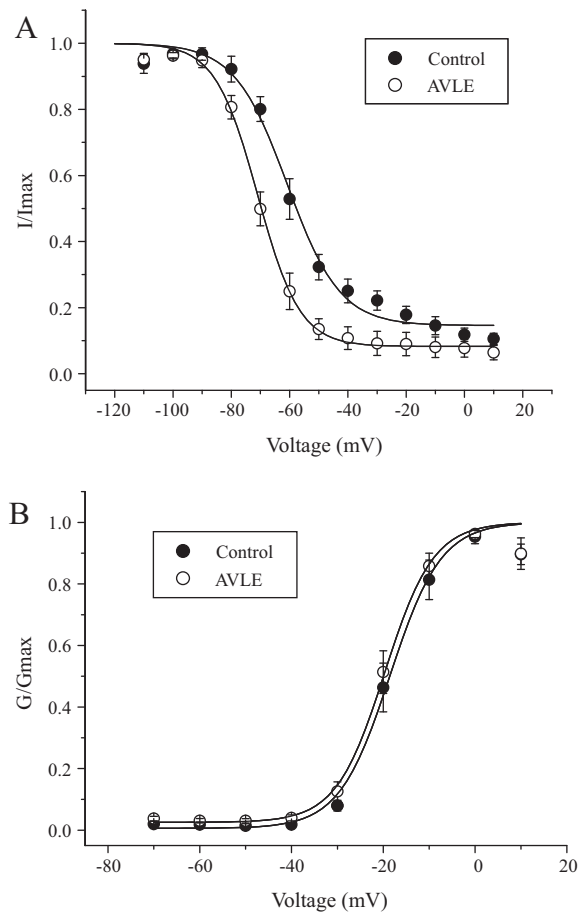


Fig. 3. AVLE caused a left-shift in steady-state inactivation but did not affect voltage-dependence of activation of Na⁺ currents in N2A cells. (A) The steady-state inactivation protocol was performed in cells treated with and without 20 μg/ml AVLE. A test pulse step of –10 mV was preceded by a pre-pulse of 2 s of different potentials. The test pulse currents are then normalized to the largest test pulse current and plotted against the pre-pulse voltages. The results are fitted by the Boltzmann equation. Results are mean ± SEM from 6 cells of each group. (B) Voltage-dependence of activation: Na⁺ currents were triggered by increasing depolarization (from a holding potential of –70 mV and then 10 mV increments), and conductance (*G*) is calculated as described in the Methods section. Each conductance in the control group and the AVLE-treatment group is then normalized with the respective maximum conductance (*G_{max}*) and then plotted against the applied depolarization voltages. The results are then fitted with the Boltzmann equation. Results are mean ± SEM from 7–9 cells of each group.

3. Results

Fig. 1A shows that addition of 20 μg/ml AVLE suppressed the Na⁺ currents (tonic block). The current–voltage relationship is summarized in Fig. 1B. The concentration-dependent inhibition curve is shown in Fig. 1C and the IC₅₀ value is calculated to be 18.4 μg/ml. The inhibition by AVLE (20 μg/ml) was rapid and readily reversed by washout (Fig. 2). There was a 97.4 ± 6.2% recovery after washout (*n* = 3).

We next examined if AVLE affected Na⁺ channel gating. In the presence of 20 μg/ml AVLE, the steady-state inactivation curve shifted to the left by 13 mV (*V_{1/2}* = –55.2 ± 3.2 mV and –67.5 ± 2.9 mV in the absence and presence of AVLE, respectively; *p* < 0.05) (Fig. 3A). Voltage-dependence of activation of Na⁺ channels was not significantly affected by 20 μg/ml AVLE (Fig. 3B).

Lidocaine, a local anesthetic, inhibits Na⁺ channels with use-dependence such that the cumulative block enhances with stimulation frequency (Hille, 2001; Leung et al., 2010a). We then investigated whether AVLE blocked Na⁺ currents with use-dependence. As shown in Fig. 4, current magnitude in the

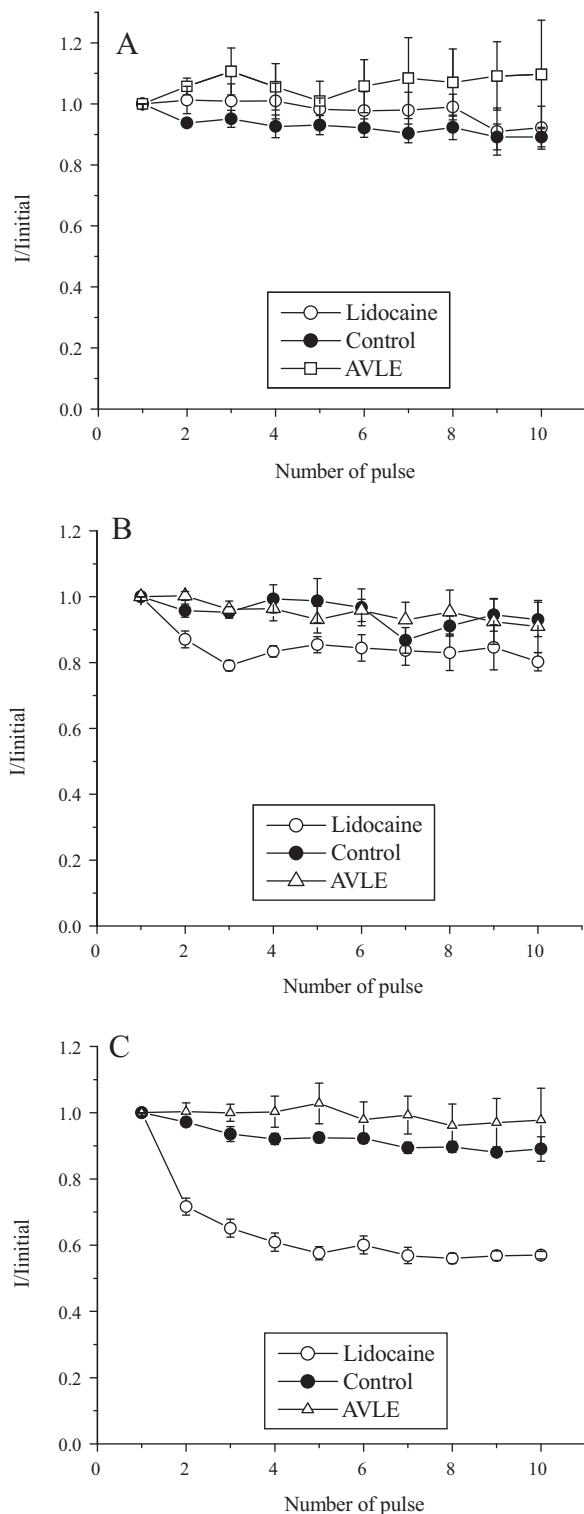


Fig. 4. AVLE block of Na^+ currents in N2A cells displayed no use-dependence. Cells were stimulated by -10 mV pulses at (A) 0.33 Hz (B) 1 Hz and (C) 3.33 Hz in the absence or presence of $20 \mu\text{g/ml}$ AVLE or $30 \mu\text{M}$ lidocaine. The maximum current amplitudes of the second to tenth pulses (I) are normalized with the maximum current amplitudes of the first pulse (I_{initial}) and then plotted against the number of pulse. Results are mean $\pm$ SEM from 3–5 cells of each group.

control was not significantly affected by repeated stimulation (0.33–3.33 Hz; Fig. 4A–C). In the presence of $30 \mu\text{M}$ lidocaine, inhibition did not increase with consecutive stimulation at 0.33 Hz, but raising the stimulation rate to 1 and 3.33 Hz enhanced the cumulative block in a frequency-dependent fashion (Fig. 4A–C). However,

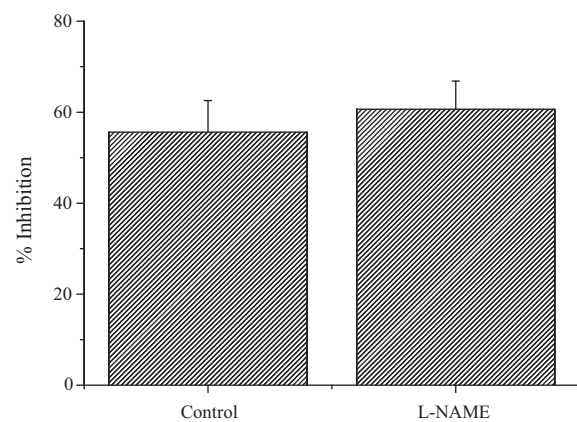


Fig. 5. AVLE block of Na^+ currents in N2A cells did not depend on NO release. Cells were pre-treated with and without $300 \mu\text{M}$ L-NAME for 10 min. After a whole-cell configuration was established, the cells were stimulated with -10 mV pulses every 10 s and AVLE ($20 \mu\text{g/ml}$) was added to the bath. AVLE inhibition of Na^+ currents was quantified in the control group and the L-NAME-pretreated group. Results are mean $\pm$ SEM from 3–5 cells of each group.

the extent of block by AVLE was not affected upon repeated stimulation at all the frequencies tested. Thus, AVLE block did not exhibit use-dependence.

Since AVLE has been shown to cause prominent vasodilatation via the stimulation of NO release from endothelial cells (Kwan et al., 2005; Tagawa et al., 2004), we examined whether the effect of AVLE on Na^+ currents could also be mediated, directly or indirectly, by

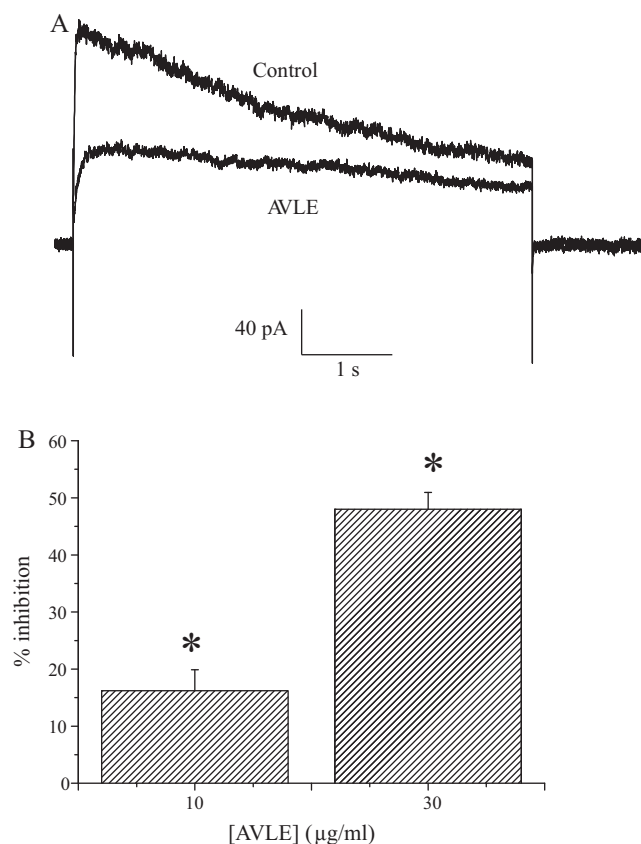


Fig. 6. AVLE inhibited voltage-gated K^+ currents in N2A cells. (A) Outward K^+ currents were triggered by $+30$ mV pulses every 20 s and AVLE ($30 \mu\text{g/ml}$) was added to the bath. The traces show the currents obtained before AVLE and after the inhibitory effect of AVLE has reached a steady-state. (B) indicates the % inhibition of voltage-gated K^+ currents by AVLE (10 and $30 \mu\text{g/ml}$). * indicates statistical significance ($p < 0.05$). Results are mean $\pm$ SEM from 4–5 cells of each group.

the release of NO. Our results show that the % block of Na^+ currents by AVLE was not significantly affected by pretreatment of L-NAME, an inhibitor of NO generation (Fig. 5). These results suggest that the blockade of Na^+ currents by AVLE was unrelated to NO release as observed in endothelial cells.

Activation of vascular smooth muscle K^+ channels and thereby promotion of K^+ efflux was proposed to be part of the mechanisms for the AVLE-triggered vasorelaxation (Kwan et al., 2005; Tagawa et al., 2004). We therefore examined whether AVLE could affect voltage-gated K^+ channels in N2A cells. Addition of 10 and 30 $\mu\text{g}/\text{ml}$ AVLE suppressed the K^+ currents by 16.2 ± 3.7 and $48.0 \pm 2.9\%$, respectively (Fig. 6).

In view of the results described above, we also investigated whether AVLE affected K_{ATP} channel currents. Since the latter channels are absent in N2A cells (Leung, unpublished observation), we used rat insulinoma RINm5F cells, which expressed robust K_{ATP} channel currents (Fig. 7). It has been known that beta cells and neurons share the same molecular composition of K_{ATP} channels (Kir6.2 and SUR1 subunits) (Aguilar-Bryan and Bryan, 1999). As shown in Fig. 7A, the low concentration of intracellular ATP (2 mM) allowed K_{ATP} channel currents (manifested as hyperpolarization-triggered inward currents) to develop slowly after break-in. Addition of

100 μM diazoxide (a K_{ATP} channel opener) accelerated channel opening. After the effect of diazoxide has reached a maximum, application of 10 μM glibenclamide (a K_{ATP} channel inhibitor) completely suppressed the K_{ATP} currents. The maximal current amplitude changes in Fig. 7A are plotted against time and shown in Fig. 7B. Fig. 7C and D shows the results of an experiment using the same protocol as described in Fig. 7A, except that 30 $\mu\text{g}/\text{ml}$ AVLE was added soon after the whole-cell configuration was established. Quantitative results are summarized and shown in Fig. 7E. AVLE on its own did not affect the slow development of K_{ATP} currents in the presence of 2 mM intracellular ATP; neither did it modulate K_{ATP} channel opening in the presence of diazoxide.

4. Discussion

Although the psychopharmacological effects of AVLE have been known and demonstrated in several animal studies (Butterweck et al., 2001, 2003; Grundmann et al., 2007, 2009), and that AVLE has been reported recently to protect neurons *in vitro* from injury by oxygen and glucose deprivation and corticosterone challenge (Xiang et al., 2010; Zheng et al., 2011), there is hitherto very lit-

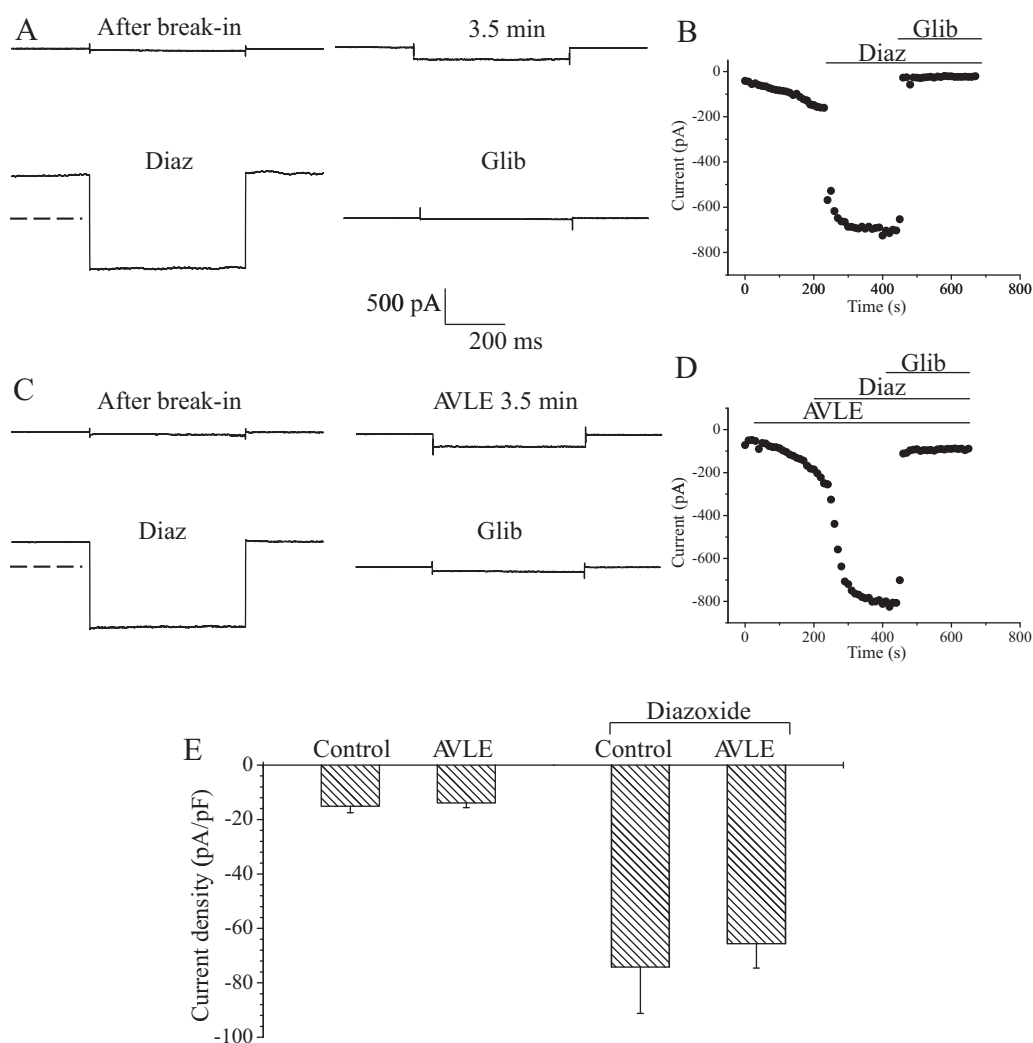


Fig. 7. AVLE had no effect on K_{ATP} channel currents in rat insulinoma RINm5F cells. (A) Traces shown represent inward currents triggered by hyperpolarizing pulses (-140 mV) in RINm5F cells dialyzed with 2 mM ATP in the pipette just after a whole-cell mode was established, 3.5 min after the whole-cell mode, after the stimulatory effect of 100 μM diazoxide (Diaz) has reached a maximum and finally after 10 μM glibenclamide (Glib) addition. The dashed line indicates the zero-current level. (B) indicates the results from (A): maximum inward currents are plotted against time. (C) and (D) show results from an experiment using the same protocol as in (A), except that AVLE (30 $\mu\text{g}/\text{ml}$) was added soon after the whole-cell configuration was established. Quantitative results are summarized in (E). Results are mean $\pm$ SEM from 6–7 cells of each group.

tle information about whether AVLE has any effect on neuronal ion channels. In the present work, we for the first time reported that AVLE concentration-dependently suppressed voltage-gated Na^+ channels in neuronal cells in a rapid and fully reversible manner.

AVLE did not affect the kinetics and voltage-dependence of activation of Na^+ channels, but caused a significant left shift in the steady-state inactivation curve (Fig. 3). The latter result suggests that in the presence of AVLE, the availability of Na^+ channels is reduced and that a higher proportion of Na^+ channels is in the inactivated state.

Local anesthetics such as lidocaine have two modes of block of Na^+ channels, namely, tonic and phasic block (Hille, 2001; Leung et al., 2010a). The former represents closed channel block in which block of channels could happen without channel opening. The latter block depends on channel activation, in a manner that the cytoplasmic activation gate opens to permit the drug to gain access to the internal vestibule. Thus, the more frequently the channels open, the higher the degree of cumulative block. We previously showed that osthonol, a herbal compound which has antihypertensive and neuroprotective effects, could exhibit both tonic and phasic block of Na^+ channels in N2A cells (Leung et al., 2010a). Our data here show that AVLE only displayed tonic block but not phasic block, since the block by AVLE was not enhanced by increasing the frequency of Na^+ channel opening. We also reported before that diphenidol, an anti-emetic agent and also a potential local anesthetic, exerted a potent tonic block but not phasic block of Na^+ channels in N2A cells (Leung et al., 2010b).

AVLE has been reported to have anxiolytic effects in rats in the elevated plus maze test, and such an anxiolytic effect was found to be caused by activation of GABA_A receptors and 5-HT_{1A} receptors (Grundmann et al., 2007, 2009). It has been proposed that Na^+ channel blockade is one of the mechanisms leading to the anti-anxiety effects of multiple drugs such as lamotrigine, carbamazepine and riluzole (Bourin et al., 2009; Mirza et al., 2005). Furthermore, blockade of Na^+ channel activities has been known to be associated with analgesic efficacy of antidepressants (Dick et al., 2007). The suppression of Na^+ currents by AVLE may thus in part contribute to the anxiolytic property of this herbal remedy. Whether the Na^+ channel-blocking effect of AVLE may endow this herbal medicine an efficacy as an analgesic would warrant future investigation.

Increasing evidence has shown that depressed patients are characterized by excessive NO production and NO synthase inhibitors are effective in treatment of depression (Galecki et al., 2010; Steinert et al., 2010). In animal models exhibiting depression-like symptoms, NO synthase inhibitors exert anti-depressant actions (Kulkarni and Dhir, 2007). AVLE has been shown to cause vasodilation via stimulation of endothelial NO release (Kwan et al., 2005; Tagawa et al., 2004). If AVLE were to generate NO in N2A cells as effectively as in vascular endothelial cells, the enhanced level of NO is not compatible with its anti-depression effect. On the other hand, the inhibition of Na^+ entry by AVLE may lead to reduced cytosolic level of Ca^{2+} (e.g., via the $\text{Na}^+/\text{Ca}^{2+}$ exchange mechanism) and thus reduced NO formation. Consequently, the decreased NO level may account for the anti-depressive effect of AVLE. Further, our results show that the % block of Na^+ currents by AVLE was not significantly affected by pretreatment of L-NAME, a potent inhibitor of NO generation by any NO synthase isoform. These results suggest that the block of Na^+ currents by AVLE was not due to NO release.

Activation of vascular smooth muscle K^+ channels and thereby promotion of K^+ efflux was proposed to be part of the mechanisms for the AVLE-triggered vasorelaxation (Kwan et al., 2005; Tagawa et al., 2004). In the latter work, since TEA or glibenclamide could suppress the effects of AVLE, we examined if AVLE activated voltage-gated K^+ channels and/or K_{ATP} channels. Our results indicate that AVLE mildly inhibited voltage-gated K^+ currents in N2A

cells. Since N2A cells do not express K_{ATP} channels (Leung, unpublished results), we studied the K_{ATP} channels in RINm5F beta-cells. Islet beta-cell and neuronal K_{ATP} channels share the same molecular compositions: inward rectifier K^+ channel (Kir6.2) and sulphonylurea receptor (SUR1). We reported here that AVLE had no effect on K_{ATP} channels: it did not on its own activate the K_{ATP} channel, nor did it affect the K_{ATP} channel opening induced by diazoxide. Therefore, results in this report and our previous report seem to suggest that AVLE has different effects on voltage-gated K^+ channels and K_{ATP} channels in neuronal cells and vascular smooth muscle cells. One of the plausible causes for these differential effects could be the expression of different K^+ channel members in neurons and smooth muscle cells. The major Kv members expressed in N2A cells are Kv1.1 and Kv2.1 (Leung et al., 2011) while Kv1.2 and Kv1.5 are the major Kv members in vascular smooth muscle (Cox et al., 2001; Thorneloe et al., 2001). K_{ATP} channels in smooth muscle cells have been known to have compositions (Kir6.2 plus SUR2B) different from those of islet beta cells and neurons (Kir6.2 and SUR1) (Aguilar-Bryan and Bryan, 1999).

Neurons beginning to undergo apoptosis have been known to over-express voltage-gated K^+ channels; the excessive K^+ efflux results in lowering of cytosolic K^+ concentration and eventually leads to relief of inhibition (hence activation) of pro-apoptotic caspases and nucleases (Yu, 2003). Voltage-gated K^+ channel blockers have therefore been considered as potential neuroprotective drugs (for a review see Leung, 2010). A recent report demonstrates that AVLE could protect rat cortical neurons from hypoxic challenge by inhibiting the activities of pro-apoptotic caspases (Xiang et al., 2010). The possibility that the neuroprotective effect of AVLE may in part result from its inhibition of neuronal Kv channels (Fig. 6) could not be ruled out.

5. Conclusions

AVLE has now been shown for the first time to cause blockade of Na^+ channels in a rapid, reversible and non-use-dependent manner in cultured neuronal cells. This finding provides a plausible explanation for the antidepressant and anxiolytic effects of AVLE in patients and experimental animals. The inhibitory effect of AVLE on voltage-gated K^+ channels, together with its antioxidant effect, may also partly account for its recently reported neuroprotective actions (Xiang et al., 2010; Zheng et al., 2011).

Conflict of interest

The authors declare no conflict of interests.

Acknowledgments

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